LoRaWAN Gate Opener

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ABSTRACT

This article describes how LoRaWAN devices can be applied and used to open a gate. The objective was to create a low-power, long range communication device for securely opening a gate. This study was tested using an Arduino Uno with a Dragino LA66 shield and a LoRaWAN gateway. Using code to send a signal from the LoRaWAN gateway to the arduino uno connected to a LA66 board. The arduino receives the signal and tells the gate to open. We found that the LoRaWAN had no issue sending the signal from several hundred feet away to the gate. These findings suggest that the long range and efficiency of the LoRaWAN can be effective in smart infrastructure projects or in optimizing everyday activity.

CCS CONCEPTS

- **Hardware** → Wireless devices
- **Computer systems organization** → Embedded and cyber-physical systems
- **Networks** → Sensor networks
- **Information systems** → Internet of Things

KEYWORDS

LoRaWAN, IoT, actuator systems, remote control, embedded systems, rural automation, TTN, wireless communication

1 Introduction

In agricultural settings, physical access to infrastructure like automatic gates or various other farm equipment can be limited by distance and time. Traditional solutions such as wired landlines are often either not feasible or outdated. This project aims to show how these problems can be overcome using LoRaWAN technology, by building a real-world example: a remote-controlled gate opener. Using the project as an example, problems like distance, obstruction and lack of WiFi coverage can be overcome.



Figure 1: The Gate to the Painter Farm

2 Background and Motivation

In our setting, the Painter household's farm, the gate to the road used to have a landline that could be used to open the gate from the house. This enabled the family to let in visitors without having to traverse the quarter of a mile road down to the gate. However, after getting rid of the landline, the ability to open the gate from the house was also cut. Now they need a new solution, and that is what this project aims to provide.

3 Existing Solutions and Our Approach

As previously discussed there exist other solutions for this problem such as landlines that connect the gate to the house, these solutions are very costly and require a lot of manual labor to install, so instead an easier and more optimal solution would be to use a long-range, low-power system like the LoRaWAN network to control the gate wirelessly. Our proposed solution integrates a heartbeat-driven LoRaWAN communication loop between an Arduino controlled relay, and a LoRaWAN gateway back at the house. When a downlink is received via the TTN network, the Arduino triggers the relay down at the gate, simulating a button press on the gate opener to open the gate. The solution avoids the need for cables or other infrastructure.

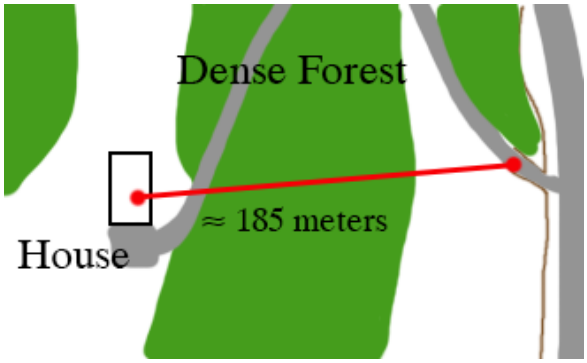


Figure 2: The distance between the farmhouse and the gate in a straight line. The distance the driveway follows is much longer.

4 Literature Review

There have been recent advancements in low power wide-area networks (LPWAN) that made technologies such as LoRaWAN more appealing for remote operating and control applications. LoRaWAN offers long-range communication with low power consumption making it ideal for rural and urban areas [2]. Several studies have applied the LoRaWAN in different applications requiring reliable data transmission highlighting the effectiveness of the device in rural and urban areas. Previous implementations of wireless control systems have relied on WiFi, bluetooth, or GSM

technologies. However, these approaches are limited by range, power consumption, and cost [1]. A recent study by Figueiredo [1] investigated how LoRaWAN performs in sporadic conditions meaning, places with a lot of disturbances and what algorithms can combat that. They tested that even within a sporadic environment the LoRaWAN performed better than the previous implementations mentioned. However, there were still limitations thus they developed an algorithm incorporating dynamic time slot allocation, load-based adjustments, prioritization of critical messages. The devices are assigned to specific time slots minimizing collision. Then the algorithm adjusts the number of time slots based on the network load. Finally, time sensitive data are given higher priority to ensure timely delivery.

This project builds upon prior research by applying LoRaWAN to remotely open a gate. Unlike prior studies focusing on sensor data collection, this study focuses on actuator control. This provides information into delay sensitivity, reliability, and security concerns. An algorithm described by Figueiredo [1] will be beneficial in implementing the gate in a city area or a place with a lot of people.

5 Implementation

5.1 Hardware

- Arduino Uno
- Dragino LA66 LoRaWAN Module
- LPS8v2 LoRaWAN Gateway
- DC 5V 1-Channel Relay Module
- LINEAR Multicode 1-Button Gate Remote 308911

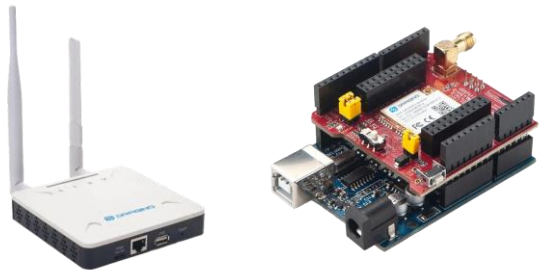


Figure 3: LPS8v2 LoRaWAN Gateway (left); Arduino Uno & Dragino LA66 LoRaWAN Module (right)

5.2 Physical Setup

In order to implement our solution, we first had to take an old gate remote and disassemble it. After taking it apart I desoldered the button from the board, and soldered in 2 wires that connected to our 5V relay. After connecting the relay to the 5V, ground, and 8 I/O pins on the LA66 board, the physical setup for our solution was pretty much complete.

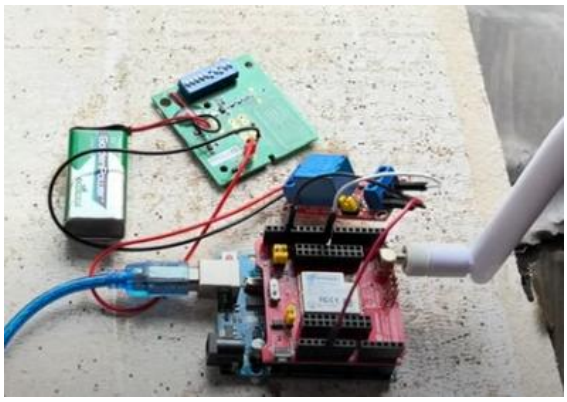


Figure 4: The physical setup for the project.

5.3 Digital Setup

The digital side of things was a bit more complicated, I needed to set up the LA66 board so it was able to connect to the TTN network. I also needed to set up the LoRaWAN Gateway so that it was connected to the local network and the TTN network as well. After getting these setup and acquiring the DevEUI, AppEUI and AppKey we were ready to connect the device to the TTN network and start to write the code. To that end, we took one of the given examples, the “Join-TTN-Network” script and retooled it a bit so that we could receive our downlinks. And once a downlink with the correct payload was received, we would send a HIGH signal to the 8 I/O pin, triggering the relay and gate opener, sending a signal out to our gate.

6 Problems and Challenges

During our implementation of the project, we ran into various problems and concerns.

6.1 Initial Issues with Receiving Downlinks from the Network

During our initial creation of the project, we had quite a lot of trouble with getting downlinks to the LA66 board, and after a lot of trial and error, we found that the TTN network was only able to schedule downlinks to the LA66 board after receiving an uplink first. To solve this issue we implemented some “heart-beat” uplinks, that would be sent once every 30 seconds to the network, and any scheduled downlinks from the server would be sent back to the LA66 board.

On top of this issue however, we ran into another big problem. For whatever reason our board wasn’t able to receive downlinks correctly. Whenever we would try to send a downlink, the resulting text would be output as a garbled message. In order to fix this we made sure to format our downlink payloads on the TTN console first. This completely fixed the issue and allowed us to move on to testing our solution.

6.2 Distance and Blockage

Before testing we had concerns that the distance between the gate and the house would be too great to overcome, and the forest that existed between the house and the gate would cause too much interference between the Gateway and the

LA66 module. However, upon testing we found that these factors seemed to be a non-issue. We didn't really have any trouble at all, establishing a link between the LA66 module and the gateway back at the house. Now this was on a fairly clear day, during the afternoon, so conditions may have been optimal, but testing the device from different distances and spots around the farm yielded similar results. On a different day, at a different time, perhaps we'd have more trouble, but for our purposes the LA66 board seemed to perform just fine.

7 Results

We conducted multiple experiments testing the connectivity of the signal at different distances for the device. After conducting several tests, we successfully opened the gate each time, confirming that the LoRaWAN system was able to perform reliably across the range of our farm.

However, while the functionality was consistent, we did identify a few areas for improvement. One noticeable issue was the delay between triggering the gate remotely and the actual execution of the command. Although our request reached the LA66 board without fail, it often took several seconds for the downlink to be delivered. This delay is likely due to the nature of our heartbeat system—downlinks from TTN can only be sent in response to an uplink, and our heartbeat was configured to send every 30 seconds. As a result, the worst-case delay before the command is received could be close to half a minute.

Despite this timing limitation, the connection held steady, and we experienced no packet loss or signal drops, even when testing beyond the line of sight. Overall, the results confirmed the reliability and range of LoRaWAN in a rural, real-world environment.

8 Future Improvements

There are plans to continue this project well after the class concludes, as it is a great solution for the Painter farm. To this end, several improvements to the project will probably be made:

8.1 A User Interface

In the future I have plans to implement a user interface of some kind, whether it's an app that users can access from their phone, or a physical button in the house that the occupants can press to open the gate.

8.2 3D Printed Enclosure

In order to keep the Arduino Uno down at the gate, it will be necessary to create some sort of weatherproof enclosure for it, so that it will not be damaged by the elements. This could either be 3D printed or some other existing enclosure.

8.3 Sensors

Currently the Arduino Uno sends an uplink to the network once every 30 seconds. While this is fine normally, by adding a sensor outside of the gate, we could detect when a car is sitting at the gate. This would let us completely stop uplinks to the server and go to sleep when a car is not at the gate, and instead increase uplink polling to once every 10 or even 5 seconds, making our solution much more responsive.

8.4 Other Projects

Due to the versatility that a LoRaWAN network offers, I can imagine using it in several other projects in the future. For example: sending a signal back to the house when a package has been delivered or monitoring the water levels of our animals' drinking water.

9 Conclusion

This project set out to solve a simple but meaningful problem: how to open a distant gate without running new infrastructure or relying on outdated systems. By applying LoRaWAN technology, we designed and implemented a low power, long range solution that reliably triggered a gate opener using wireless communication. Through this hands-on process, we tackled real world challenges like downlink timing, signal stability, and environmental interference.

Beyond solving a problem on the Painter farm, this project demonstrated how LoRaWAN can be used for actuator control rather than just data collection. This opens up new possibilities for smart infrastructure in remote or resource-constrained settings. It showed that with affordable components and open network tools, it is possible to build practical IoT solutions that are both robust and scalable.

ACKNOWLEDGMENTS

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